

Working today



- 23 plugin libraries, composites, wizards
- GA calibration and MCMC with joint error estimation
- Model compiler, 5–15× faster solves
- Headless runners, Python bindings, Docker
- OpenHydroTwin: live forcing, streaming assimilation, browser viewer

In flight



- Streaming MCMC in the twin, so the live loop carries a full posterior and not just a point estimate
- Online drift detection for asset condition
- Running the twin's assimilation on a compiled kernel
- Web-based model setup and visualisation

Planned



- Real-time control and operational optimisation
- Machine-learning integration alongside the process model
- Analytical Jacobian by differentiating the expression trees
- Auto-generated plugin documentation
- Portfolio-scale deployment by configuration alone

Everything on the left is in the repository today and is what this talk has shown. The middle column is being built now against the bioretention and wetland test beds. The right column is where the platform is headed, and is the part most open to what a partner actually needs.